

Residual Industry Momentum

Uma Malani

Daniel McCance

PGI Quant Education — Spring 2026

Research Question and Motivation

Industry momentum (Moskowitz and Grinblatt, 1999) is a robust empirical anomaly: portfolios long industries with high trailing 12-month returns and short industries with low trailing returns earn positive risk-adjusted returns. However, raw industry returns are a noisy signal of true industry-specific momentum because they conflate idiosyncratic industry effects with passive exposure to common factors. When the market rallies, high-beta industries appear as “winners” due to their factor loadings rather than their industry-specific outperformance. We hypothesize that stripping out Fama-French 3-factor exposures before computing momentum will isolate the industry-specific underreaction signal, producing a strategy with similar or higher returns, lower volatility, and meaningfully reduced exposure to the standard momentum factor.

Literature Review and Contribution

Moskowitz and Grinblatt (1999) established industry momentum as the dominant formulation of the broader Jegadeesh-Titman (1993) momentum anomaly, showing that controlling for industry effects largely absorbs stock-level momentum. Blitz, Huij, and Martens (2011) applied factor-residualization to individual stock momentum and found that the resulting “residual momentum” preserves average returns while approximately halving volatility, with materially reduced exposure to momentum crashes. Daniel and Moskowitz (2016) further characterized these crashes as predictable consequences of factor exposure reversing in bear-market rebounds. Our contribution applies the Blitz-Huij-Martens residualization framework to the industry-level momentum of Moskowitz-Grinblatt.

Method

We pull 49 value-weighted industry portfolios and the FF3 + UMD factors from Ken French’s data library (July 1963 through April 2026). For each industry at each month-end, we compute two signals: a baseline 12-1 cumulative return, and a 12-1 cumulative residual return where residuals are obtained from a rolling 36-month regression of industry returns on Mkt-RF, SMB, and HML. We form equal-weighted long-short portfolios from the top and bottom 5 industries by signal, hold for one month, and compute strategy returns over a common period (1967–2026, 706 months). We evaluate via annualized Sharpe ratio, maximum drawdown, and time-series regression on the FF3 + UMD factor model with Newey-West (HAC, 6 lags) standard errors. The alpha intercept and its t -statistic are our primary metric. We test robustness across three momentum lookbacks (6, 9, 12 months), three residualization windows (24, 36, 48 months), and two factor models (CAPM and FF3), yielding 21 total variants.

Results

The baseline 12-1 industry momentum strategy earns 10.9% per year with 21.0% volatility (Sharpe 0.52), but its alpha after factor controls is not statistically significant ($t = 1.31$). The strategy’s factor loadings reveal it is essentially a tilted version of the standard momentum factor—UMD beta

of 1.06, factor model R^2 of 0.53. Residual industry momentum earns 9.2% per year with 16.2% volatility (Sharpe 0.57), but produces a statistically significant alpha of 4.4% annualized ($t = 2.15$). Critically, the residualization halves both the UMD loading (to 0.54) and the model R^2 (to 0.22), indicating that the residual signal captures information not already spanned by standard factors. The two strategies also differ qualitatively during crisis periods: the baseline strategy suffers the classic Daniel-Moskowitz momentum crash in March-April 2009 (losing approximately 38% in two months), while the residual strategy declines more gradually through 2008 without the concentrated cliff event.

Robustness

At the 12-1 lookback, all six tested parameter combinations (varying residualization window and factor model) produce statistically significant alpha, with t -statistics ranging from 2.15 to 4.25 and Sharpe ratios from 0.57 to 0.75. The strongest variant uses a shorter 24-month residualization window with full FF3 residualization (Sharpe 0.74, alpha t -statistic 4.25, maximum drawdown -40% versus baseline -58%). At shorter lookbacks (6-1 and 9-1), residualization does not produce significant alpha. We interpret this as consistent with the underlying economic mechanism: 12-month momentum reflects investor underreaction at a horizon where information has fully diffused but not yet fully repriced; residualization sharpens this signal but does not manufacture it where it is absent at shorter horizons.

Limitations

All testing is in-sample on the full available history. Transaction costs are not modeled, though industry-level long-short strategies have relatively modest turnover. Analysis is restricted to US industries, so cross-validation across countries or asset classes is not performed. The joint hypothesis problem applies: we cannot fully separate “residual momentum is mispricing” from “residual momentum is exposure to an unmeasured risk factor,” and indeed Blitz-Huij-Martens have argued the residual momentum premium may reflect compensation for residual industry-specific risk that is genuinely undiversifiable.

References

- Blitz, D., Huij, J., and Martens, M. (2011). Residual Momentum. *Journal of Empirical Finance*, 18(3), 506–521.
- Daniel, K. and Moskowitz, T. J. (2016). Momentum Crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Jegadeesh, N. and Titman, S. (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *Journal of Finance*, 48(1), 65–91.
- Moskowitz, T. J. and Grinblatt, M. (1999). Do Industries Explain Momentum? *Journal of Finance*, 54(4), 1249–1290.